# Tinhinane Amina Azzouz

Curriculum vitae

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**Assistant Professor of Mathematics**
Beijing Institute of Mathematical Sciences and Applications (BIMSA)
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## Research interests

$p$-adic differential equations; non-Archimedean spectral theory; ultrametric functional analysis; Berkovich analytic geometry.

## Academic appointments

Jan. 2024–present — **Assistant Professor**
Beijing Institute of Mathematical Sciences and Applications, Beijing, China.

2021–2023 — **Postdoctoral Researcher**
Yau Mathematical Sciences Center, Tsinghua University / BIMSA, Beijing, China.

Nov. 2018–Dec. 2020 — **Assistant Professor**
Université Alger 1, Algiers, Algeria.

Sept.–Nov. 2018 — **Postdoctoral Researcher**
Max Planck Institute for Mathematics, Bonn, Germany.

Oct. 2017–Aug. 2018 — **Temporary Teaching and Research Assistant (ATER)**
Université Grenoble Alpes, Grenoble, France.

## Education

2014–2018 — **Ph.D. in Mathematics**
Université de Montpellier, Institut Montpelliérain Alexander Grothendieck, France.
Thesis: *Spectrum of p-adic differential equations.*
Supervisors: Andrea Pulita and Jérôme Poineau. Defended 11 June 2018.

2012–2014 — **M.Sc. in Mathematics and Applications**
Fundamental Mathematics, Université Pierre et Marie Curie, Paris, France. With honours.

2008–2011 — **B.Sc. in Mathematics and Computer Science**
Algebra and Cryptography, Université des Sciences et de la Technologie Houari Boumediene (USTHB), Algiers, Algeria. With honours.

## Funding and awards

2023 — **Research Fund for International Young Scientists**, National Natural Science Foundation of China; YMSC, Tsinghua University.

2014–2017 — **Funded doctoral position with teaching**, Université de Montpellier.

2012–2014 — **Paris Graduate School of Mathematics international programme**, laureate; Fondation Sciences Mathématiques de Paris.

## Academic service

Nov. 2019–June 2020 — **Deputy Director in charge of Teaching**, Department of Mathematics and Computer Science, Université Alger 1.

## Publications

1. Tinhinane Amina Azzouz. *Spectrum of p-adic linear differential equations II: Variation of the spectrum.* **Journal of the London Mathematical Society 114**(4) (2026), e70718. doi:10.1112/jlms.70718. arXiv:2303.06014.
2. Tinhinane Amina Azzouz. *Spectrum of p-adic linear differential equations I: The shape of the spectrum.* **Selecta Mathematica 30**, article 13 (2024). doi:10.1007/s00029-023-00904-4. arXiv:2111.03548.
3. Tinhinane Amina Azzouz. *Spectrum of a linear differential equation over a field of formal power series.* **Journal of Number Theory 231** (2022), 139–157. doi:10.1016/j.jnt.2020.11.021. arXiv:1808.02382.
4. Tinhinane Amina Azzouz. *Spectrum of a linear differential equation with constant coefficients.* **Mathematische Zeitschrift 296** (2020), 1613–1644. doi:10.1007/s00209-020-02482-z. arXiv:1802.07306.

## Preprints

Tinhinane Amina Azzouz, Mainak Ghosh and Sebastien Palcoux. *Local Ocneanu rigidity for separable algebra objects* (2026). arXiv:2609.10440.

## Work in progress

- Wild finite étale coverings of $p$-adic annuli, with Andrea Pulita.
- Spectral methods and decompositions for $p$-adic differential modules.
- Techniques for computing the spectrum of an ultrametric operator.

## Teaching

### Courses at BIMSA

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|---|---|
| Oct. 2026–Jan. 2027 | **Ramification Theory of Local Fields**. |
| Oct.–Dec. 2025 | **Introduction to Berkovich spaces**. |
| Feb.–May 2025 | **Introduction to $p$-adic differential equations III**. |
| Oct. 2024–Jan. 2025 | **Introduction to $p$-adic differential equations II**. |
| Mar.–June 2024 | **Introduction to $p$-adic differential equations**. |

### Earlier teaching

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|---|---|
| Aug. 2022 | **Introduction to $p$-adic differential equations**, Sanya Mathematics Summer School, Tsinghua Sanya International Mathematics Forum, Sanya, China. |
| Oct. 2015–Aug. 2016 | Teaching assignment, Université Grenoble Alpes, France. |
| Oct. 2014–Aug. 2015 | Teaching assignment, Université de Montpellier, France. |

## Selected talks

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| 14 Aug. 2026 | *The spectrum of p-adic linear differential equations.* Invited lecture, International Congress of Basic Science (ICBS), Beijing, China. |
| 6 & 13 Nov. 2025 | *The variation of the radii of convergence of linear p-adic differential equation.* Number Theory Lunch Seminar, BIMSA, Beijing. |
| 27 May 2024 | *p-adic linear differential equations.* BIMSA Member Seminar, Beijing. |
| 27 Sept. 2023 | *p-adic differential equations and radii of convergence.* Number Theory Lunch Seminar, BIMSA, Beijing. |
| 4 July 2023 | *Spectrum of p-adic linear differential equations.* Tenth Congress of Romanian Mathematicians, Romania. |
| 18 May 2023 | *The spectral theory in the sense of Berkovich.* Number Theory Lunch Seminar, BIMSA, Beijing. |
| 28 Oct. 2022 | *Finite morphism and radii of convergence of a p-adic differential equation.* Number Theory Lunch Seminar, BIMSA, Beijing. |
| June 2022 | *Spectrum of p-adic differential equations.* BIMSA–YMSC Tsinghua Number Theory Seminar, Beijing. |
| July 2017 | *Spectral theory and ultrametric differential equations.* Students' Conference on Tropical and Non-Archimedean Geometry, University of Regensburg, Germany. |
| June 2016 | *Spectre d'un système différentiel aux coefficients constants.* École jeunes chercheurs en théorie des nombres, Clermont-Ferrand, France. |
| 2014–2018 | Five talks in Ph.D. students' seminars. |

## Conference and school participation

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|---|---|
| July 2019 | *Perfectoids*, Centre Henri Lebesgue, Rennes, France. |
| July 2018 | *Berkovich Spaces 30 Years*, Institut Henri Poincaré, Paris, France. |
| July 2017 | *Berkovich Spaces, Tropical Geometry and Model Theory*, Universidad de los Andes, Bogotá, Colombia. |
| Mar. 2017 | *p-adic Analytic Geometry and Differential Equations*, CIRM, Marseille, France. |
| Aug. 2015 | Students' Conference on Tropical and Non-Archimedean Geometry, University of Regensburg, Germany. |
| June 2015 | *Géométrie et arithmétique sur les corps locaux et globaux*, Université de Caen, France. |
| June 2015 | Conference in honour of Alexandre Grothendieck, Université de Montpellier, France. |
| June 2014 | *Special Week: Berkovich Spaces and Applications*, Université de Strasbourg, France. |

## Languages

Arabic (native); French (fluent); English (fluent).